

MADISON, SD, USA

WMO: 720624

Lat: 44.016N

Lon: 97.086W

Elev: 523

StdP: 95.20

Time Zone: -6.00 (NAC)

Period: 09-19

WBAN: 00211

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.0	-22.2	-27.3	0.3	-23.7	-24.1	0.5	-21.6	15.5	-6.9	14.2	-8.1	4.5	320	0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	31.3	23.6	29.1	22.6	27.9	21.8	26.2	28.6	24.9	27.2	23.8	26.3	6.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.3	21.8	26.8	24.0	20.2	25.7	22.8	18.7	24.9	84.8	28.3	79.3	27.3	74.8	26.4	32.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.0	11.7	10.6	DB	-29.8	33.4	2.9	1.3	-31.9	34.4	-33.6	35.2	-35.3	35.9	-37.4	36.9	(4)
(5)			WB	-29.2	28.8	3.5	1.3	-31.8	29.8	-33.8	30.6	-35.8	31.3	-38.3	32.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.7	-9.3	-9.1	0.0	6.4	13.5	19.4	21.4	19.7	16.3	7.9	0.4	-7.2	(6)	
	DBStd	12.38	7.28	6.78	7.44	5.79	5.21	3.63	3.30	3.24	4.29	5.16	6.15	6.56	(7)	
	HDD10.0	2540	598	534	320	134	26	0	0	0	4	102	290	532	(8)	
	HDD18.3	4518	857	767	570	358	165	30	9	23	87	323	538	790	(9)	
	CDD10.0	1343	0	0	8	27	135	283	354	302	194	38	2	0	(10)	
	CDD18.3	279	0	0	1	1	16	62	106	66	27	1	0	0	(11)	
	CDH23.3	2403	0	0	4	22	188	547	929	491	204	17	0	0	(12)	
	CDH26.7	673	0	0	1	6	54	158	284	127	41	3	0	0	(13)	
(14) Wind	WSAvg	5.1	5.6	5.5	5.6	6.1	5.6	4.8	4.0	3.8	4.5	5.2	5.4	5.4	(14)	
(15) Precipitation	PrecAvg	677	15	17	31	64	89	117	90	78	75	55	26	17	(15)	
	PrecMax	904	32	38	89	152	150	242	172	150	185	148	80	47	(16)	
	PrecMin	473	0	4	5	15	34	46	15	17	10	5	0	1	(17)	
	PrecStd	106	9	9	21	32	38	56	45	37	46	41	22	14	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.4	12.8	22.8	27.0	31.4	32.7	33.0	32.3	30.0	26.0	18.0	9.9	(19)	
		MCWB	4.0	9.4	13.5	15.6	18.6	22.9	25.3	24.7	22.0	16.8	11.6	7.2	(20)	
	2%	DB	3.6	7.1	17.7	22.2	27.7	30.1	31.4	29.1	27.7	22.3	14.0	5.9	(21)	
		MCWB	2.4	4.7	12.1	12.9	17.9	21.9	24.7	23.8	21.1	14.7	8.4	3.2	(22)	
	5%	DB	2.2	2.9	13.9	18.7	25.0	28.0	29.7	27.8	25.9	18.9	11.4	2.8	(23)	
		MCWB	1.6	2.1	10.2	11.1	16.6	20.8	24.2	23.0	19.8	13.0	7.6	1.5	(24)	
	10%	DB	0.2	0.2	10.1	15.9	22.6	27.1	27.8	26.2	23.8	17.0	8.7	1.1	(25)	
		MCWB	0.0	-0.2	7.7	9.8	15.7	20.5	22.9	21.7	19.6	12.0	6.1	0.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.8	10.0	17.1	17.1	22.6	27.2	28.9	27.3	25.0	17.8	12.1	7.7	(27)	
		MCDB	7.2	11.8	19.3	20.9	24.9	28.7	30.2	28.1	26.7	21.1	16.8	8.6	(28)	
	2%	WB	2.7	4.5	12.9	14.8	20.1	24.7	27.3	25.2	23.4	15.9	9.8	3.5	(29)	
		MCDB	3.5	5.8	15.9	19.5	24.3	27.3	28.5	27.4	25.4	20.1	13.5	5.2	(30)	
	5%	WB	1.3	2.3	10.2	12.9	18.8	23.1	25.2	24.2	22.3	14.2	7.9	2.1	(31)	
		MCDB	1.9	3.4	14.3	16.3	23.6	25.9	27.5	26.6	24.3	17.9	10.1	3.2	(32)	
	10%	WB	0.1	0.1	7.6	11.2	17.4	21.8	24.1	23.0	20.2	12.4	6.1	0.2	(33)	
		MCDB	0.4	0.6	10.0	15.4	20.1	24.9	26.7	24.8	22.8	15.6	8.9	1.0	(34)	
(35) Mean Daily Temperature Range	MDBR		8.7	9.3	9.6	10.8	11.1	10.5	10.8	10.6	11.2	10.6	9.9	8.1	(35)	
	5% DB	MCDBR	10.3	11.3	15.8	16.7	14.6	12.6	11.8	11.8	13.1	14.9	13.7	10.5	(36)	
		MCWBR	9.1	9.0	10.1	9.0	7.3	6.6	6.4	6.9	6.7	8.3	8.3	8.4	(37)	
	5% WB	MCDBR	10.0	11.2	15.2	13.0	11.9	10.7	10.4	10.0	10.8	12.9	12.0	10.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.269	0.284	0.313	0.364	0.378	0.388	0.393	0.397	0.350	0.320	0.287	0.267	(40)	
	taud		2.423	2.404	2.413	2.295	2.339	2.367	2.347	2.317	2.429	2.486	2.486	2.442	(41)	
	Ebn at Noon		870	915	928	898	893	882	872	856	876	862	841	836	(42)	
	Edh at Noon		73	89	101	124	123	120	121	120	99	82	69	65	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.88	2.93	3.83	4.86	5.64	6.30	6.71	5.54	4.39	2.94	2.00	1.48	(44)	
	RadStd		0.18	0.24	0.34	0.44	0.41	0.43	0.24	0.43	0.33	0.35	0.17	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (37 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page